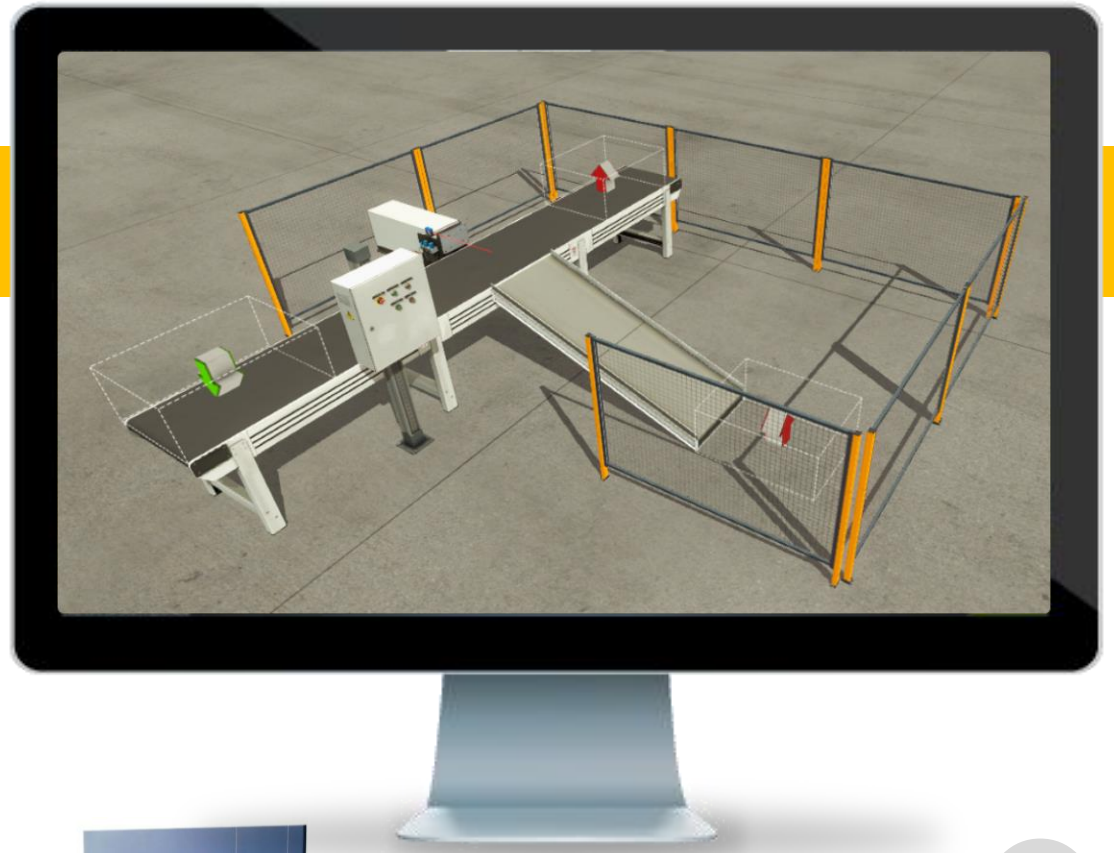
Siemens S7-1200

CPU 1212C AC/DC/Relay

Counters Application



Objective

Counting the smaller and bigger boxes

Commands to practice:

- CTU

This example is in continuation with Example 1 of Bit Logic Instructions



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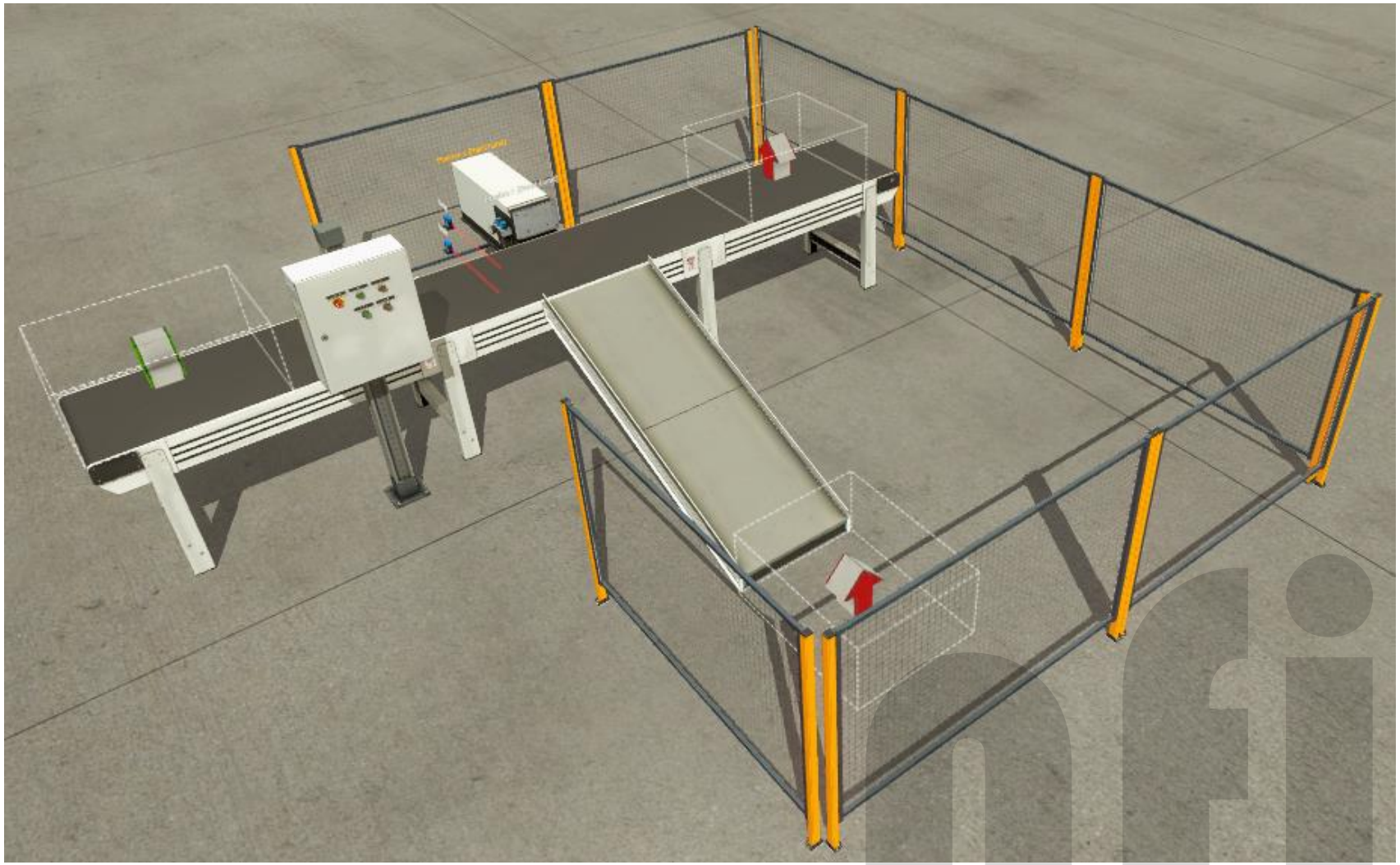
PLC used
S7-1200

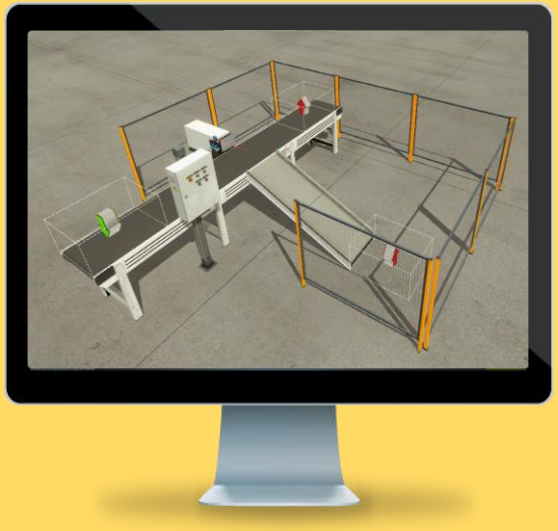
Programming Software
Siemens TIA

3D Software Platform
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Objective: Counting the small & large boxes





PLC used
S7-1200

Programming Software
Siemens TIA

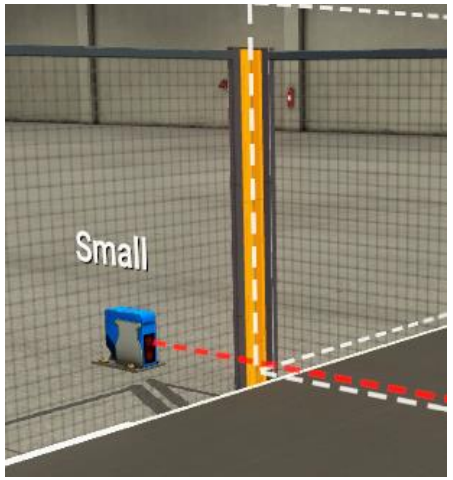
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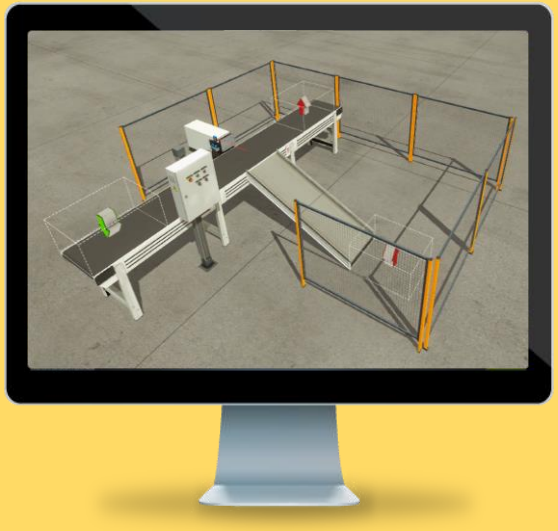
Objective: Counting the small & large boxes

Assigned Inputs & Outputs:





*Diffuse Sensor at **I0.3** has been renamed to **Large** and one sensor is added at **I0.6** for detecting **small boxes.** Reset Button **I0.7** is added to **reset the count***



PLC used
S7-1200

Programming Software
Siemens TIA

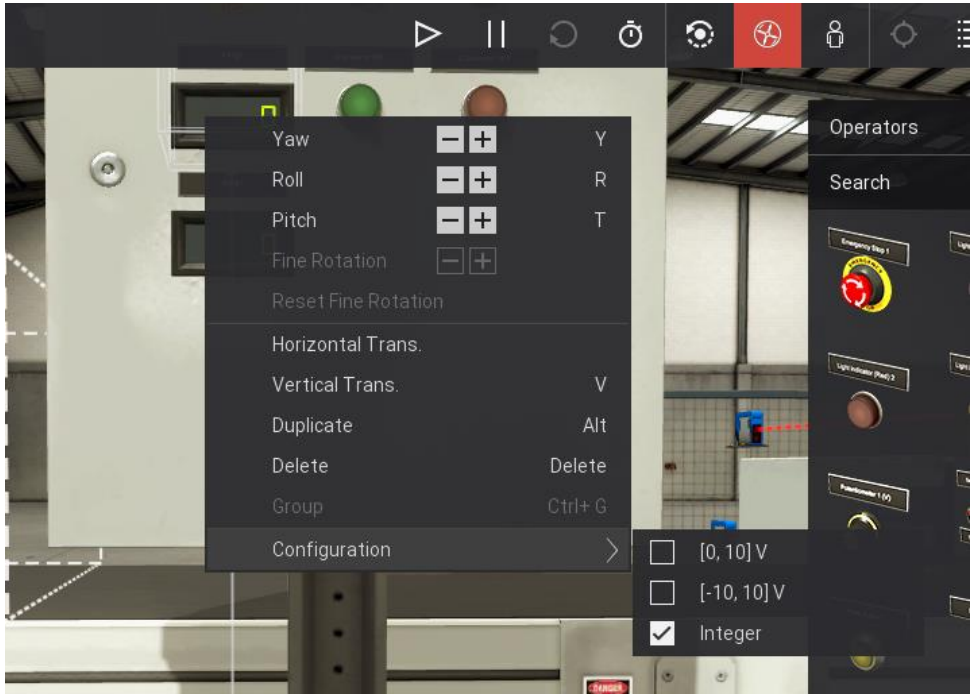
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Objective: Counting the small & large boxes

Steps to follow: Define display as Integer

Right Click -> Configuration ->Integer



Tags added in the Example 1

13	Small	Default tag table	Bool	%I0.6
14	Large Box Display	Default tag table	UDInt	%QD100
15	Small Box Display	Default tag table	UDInt	%QD104
16	Reset	Default tag table	Bool	%I0.7



PLC used
S7-1200

Programming Software
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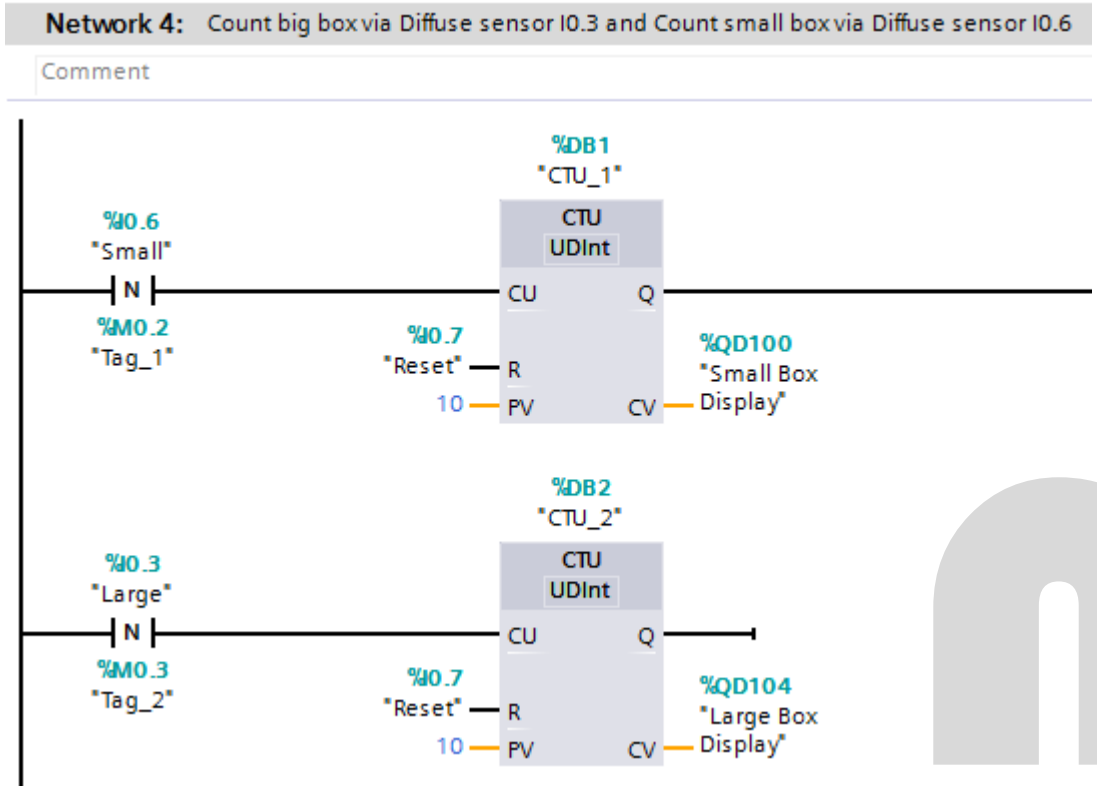
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Objective: Counting the small & large boxes

Steps to follow:

1. Open the Logic made in Example 1 – Bit Logic Instructions
2. **Count and display QD100 big box via Diffuse sensor I0.3 with reset I0.7**
3. **Count and display QD104 small box via Diffuse sensor I0.6 with reset I0.7**
4. Stop the conveyor when we have at least 10 small and large boxes
5. Download the Logic and Test!





PLC used
S7-1200

Programming Software
Siemens TIA

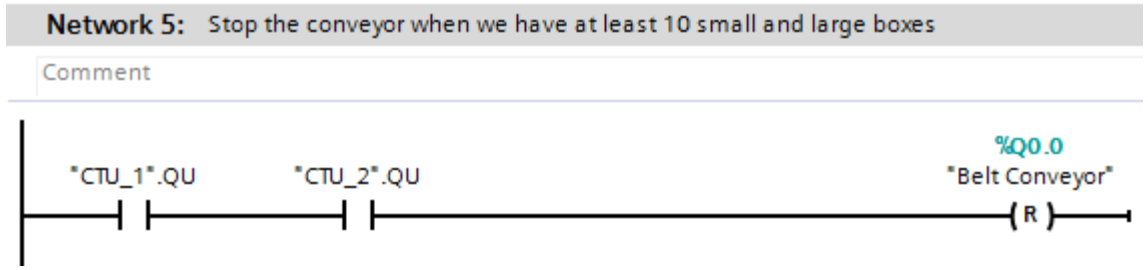
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Objective: Counting the small & large boxes

Steps to follow:

1. Open the Logic made in Example 1 – Bit Logic Instructions
2. Count and display big box via Diffuse sensor I0.3
3. Count and display small box via Diffuse sensor I0.6
4. **Stop the conveyor Q0.0 when we have at least 10 small and large boxes**
5. Download the Logic and Test!



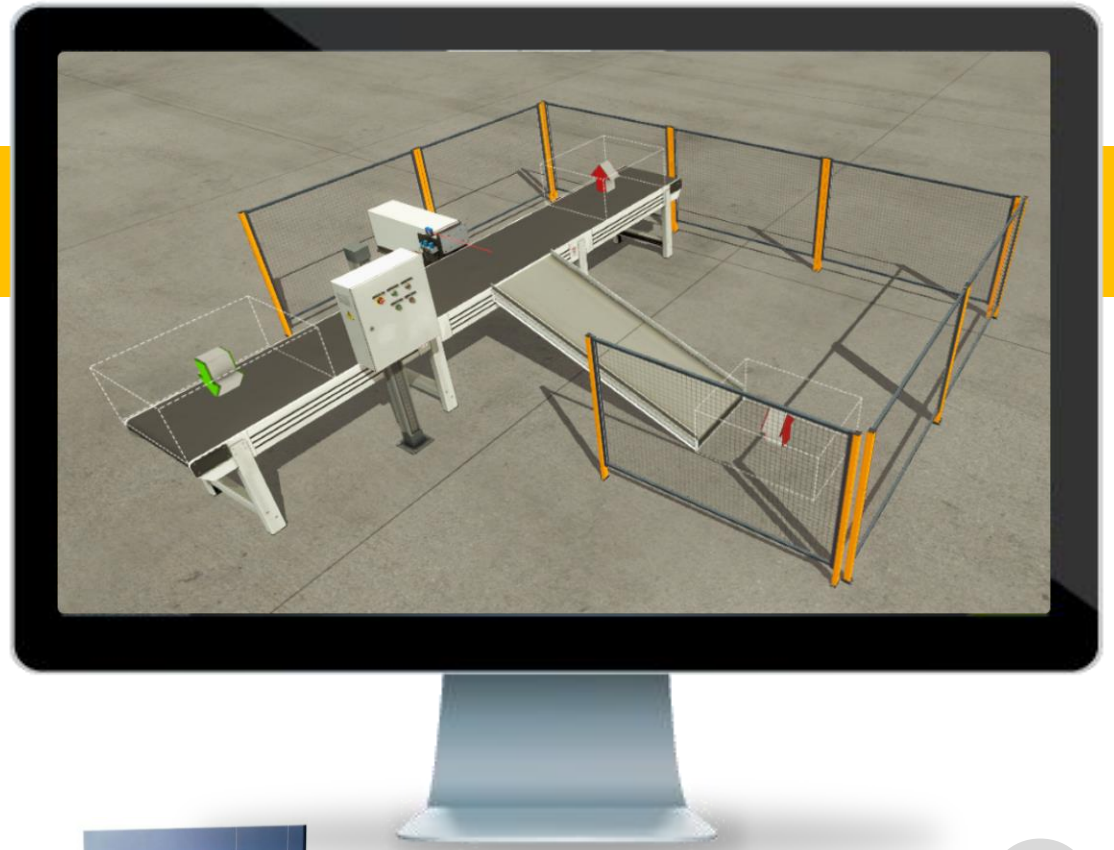
DID YOU KNOW ?

*You can control the **FACTORY I/O** environment without using hardware PLC via Control I/O Driver. This driver is available at NFI website www.nfiautomation.org. Special offer for student license.*

Siemens S7-1200

CPU 1212C AC/DC/Relay

Counter Application



Thank you

Get copy of this presentation and PLC code in the course!



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